Homework 2

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1. Question 1

1.a.

Yes, see reasoning in 1b.

1.b.

Define the payoff as X, price vector as P, and price of contingent claim as p. Solve the linear equation $Xp = P$ by elimination, we got

$$[X, P] \longrightarrow \begin{bmatrix} 1 & 0 & 0 & 0.2 \\ 0 & 1 & 0 & 0.4 \\ 0 & 0 & 1 & 0.2 \end{bmatrix} \tag{1}$$

Thus, we have complete markets in this economy and the price of continguent claim for state of nature 1, 2,3 are $[0.2, 0.4, 0.2]'$

1.c.

The price should be

$$[30, 20, 10] \times [0.2, 0.4, 0.2]' = 16 \tag{2}$$

1.d.

SDP in this economy is calucalteed by $M(i) = \frac{p(i)}{\pi(i)}$

$$M = [0.8, 0.8, 0.8]'. \tag{3}$$

This economy satisfies no-arbitrage as all the price of contingent claims are strictly positive. SDF remains the same accross all states, which means that agents in this economy are risk neutral.

1.e.

$$P_D = \sum_i \pi(i) M(i) X(i) = \frac{1}{4} \cdot 0.8 \cdot 30 + \frac{1}{2} \cdot 0.8 \cdot 20 + \frac{1}{4} \cdot 0.8 \cdot 10 = 16 \tag{4}$$

Homework
Solution

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